

How Ks are Addressed through the Project

	Ks Attributes	How Ks are Addressed through the Project	PLOs
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline	The project applies embedded systems fundamentals, UART communication, PWM motor control, RGB565 image processing, and real-time robotic navigation principles to implement autonomous following behavior using microcontroller-based hardware.	PO1
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for accepted practice areas in the engineering discipline	Specialized knowledge in embedded vision systems, robotic control architecture, serial communication protocols, and lightweight computer vision algorithms was utilized to develop the ESP32-CAM based autonomous following robot.s	PO4, PO5
WK5	Knowledge that supports engineering design in a practice area	The project emphasizes low-cost, computationally efficient, and energy-conscious robotic system design by eliminating dependency on high-performance processors, ROS, OpenCV, and deep learning frameworks while maintaining real-time performance.	PO3, PO7
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	Practical engineering technologies including ESP32-CAM, Arduino Uno, L298N motor driver, UART communication, PWM motor actuation, and embedded firmware development were integrated to construct and experimentally validate the robotic prototype.	PO5
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice	The project addresses societal and industrial challenges by reducing repetitive manual labor in retail and warehouse environments and improving user convenience through affordable robotic assistance systems.	PO6, PO7
WK8	Engagement with selected knowledge in current research literature and awareness of critical thinking and creative approaches	Existing research on autonomous robots, visual servoing, embedded robotics, and color-based tracking systems was analyzed to develop a novel dual-color fiducial marker validation strategy with improved false-positive resistance.	PO2, PO12

How Ps are Addressed through the Project

	Ps Attribute	How Ps are Addressed through the Project	PLOs
WP1	Depth of knowledge required	The project required in-depth engineering knowledge in embedded systems, robotic navigation, computer vision, serial communication, and real-time control systems to implement reliable autonomous following behavior on resource-constrained hardware.	PO1
WP2	Range of conflicting requirements	The system design involved balancing multiple conflicting requirements including low computational cost, real-time performance, detection accuracy, affordability, power efficiency, and hardware simplicity.	PO3, PO11
WP3	Depth of analysis required	The project required analytical and creative problem solving to formulate the dual-color marker validation algorithm capable of reducing false positive detections under limited processing resources.	PO2, PO7
WP4	Familiarity of issues	The implementation addressed comparatively novel challenges involving lightweight embedded vision-based tracking without OpenCV, ROS, machine learning, or external computing platforms.	PO4, PO12
WP6	Extent of stakeholder involvement and conflicting requirements	The system was designed considering practical needs of retail customers, warehouse employees, and small businesses requiring affordable autonomous assistance solutions with minimal maintenance complexity.	PO6, PO10
WP7	Interdependence	The project integrates multiple interdependent subsystems including camera-based vision processing, UART communication, embedded decision-making, motor control, power management, and robotic chassis navigation within a unified autonomous architecture.	PO5, PO9

How As are Addressed through the Project

	As Attributes	How As are Addressed through the Project	PLOs
EA1	Range of resources	The project utilized diverse resources including embedded hardware, real-time image data, low-cost financial components, robotic platforms, industry visits to shopping malls, stakeholder surveys, and engineering tools such as Arduino IDE, embedded programming, UART communication, and image processing techniques.	PO5, PO11
EA2	Level of interaction	The autonomous following robot required coordinated interaction among vision processing, embedded communication, motor actuation, power regulation, and real-time control subsystems to achieve stable operation.	PO9, PO10
EA3	Innovation	The project introduced an innovative dual-color fiducial marker and lightweight two-stage centroid-based validation algorithm capable of eliminating false positive detections without computationally intensive frameworks.	PO3
EA4	Consequences to society and the environment	The proposed robotic assistant contributes toward reducing physical workload, improving operational efficiency, and enabling affordable automation solutions for retail and warehouse environments.	PO6, PO7
EA5	Familiarity	The project extended to standard educational robotic implementations by applying principle-based embedded vision techniques and real-time system integration to solve practical autonomous following challenges.	PO2, PO12